

Figure 4: Surrogate Controls and Betti-Number Reporting

A theory-aligned control hierarchy for testing whether the observed toroidal topology is driven by oscillatory modes, aperiodic autocorrelation, amplitude statistics, phase continuity, or the reconstruction model.

Table 1. Recommended surrogate hierarchy

Condition	Preserves	Destroys / perturbs	Null hypothesis / scientific meaning	Expected Betti-number shift
Real LFP	All spectral, amplitude, phase, burst, and nonstationary structure.	Nothing.	Observed reference distribution against which all null ensembles are compared.	Enrichment of the two-torus signature $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$.
Fourier phase shuffle (current control)	Fourier magnitude spectrum and therefore the periodogram/PSD.	Fourier phase organization; temporal waveform structure; amplitude histogram is generally Gaussianized.	Tests whether the observed topology exceeds what is produced by the power spectrum alone.	Reduced $(1, 2, 1)$; increased partial or other configurations.
IAAFT surrogate	Approximate PSD/autocorrelation and the exact empirical sample-value distribution.	Organized Fourier-phase relationships and nonlinear temporal dependence beyond a static amplitude transform.	Tests whether topology is fully explained by linear second-order structure plus the non-Gaussian amplitude distribution.	If real > IAAFT, full torus detection requires temporal phase organization beyond PSD and amplitude statistics.
Aperiodic-only 1/f surrogate	Aperiodic slope/offset, broad autocorrelation, variance, duration, and optionally the empirical amplitude histogram.	Narrowband oscillatory peaks and their corresponding near-unit complex pole pairs.	Tests the manuscript prediction that aperiodic structure thickens the manifold but does not add circular factors.	Very little $(1, 2, 1)$; mostly no-loop, weak-loop, or unresolved configurations.
Envelope-preserving phase reset	Peak frequencies, oscillatory power, amplitude envelopes, burst timing, nonstationarity, and broadband residual.	Continuous phase evolution across bursts or tapered blocks; long-range traversal of independent phase coordinates.	Tests whether oscillatory bursts alone are sufficient, or whether sustained phase continuity is required for a robust torus.	Shift from full $(1, 2, 1)$ toward partial $(1, 1, 1)$, one-loop, or other classes.
One-mode ablation (mechanistic positive control)	One dominant oscillatory mode, broadband residual, and as much of the original amplitude modulation as possible.	One selected oscillatory mode / one conjugate pole pair.	Directly tests the discrete theory that each sustained oscillatory mode contributes one independent circular factor.	Predicted transition: $(1, 2, 1) \rightarrow (1, 1, 0)$. Removing both modes should resemble the aperiodic-only null.
Constrained non-oscillatory AR/TVAR (optional)	Local autocorrelation, variance, slowly varying coefficients, and window-specific nonstationarity.	Near-unit complex pole pairs; sustained oscillatory factors.	Tests whether generic locally stationary AR dependence can create false toroidal topology without oscillatory poles.	Low full-torus rate despite realistic local autocorrelation and coefficient drift.

Recommended primary package: IAAFT + aperiodic-only 1/f + envelope-preserving phase reset. Add one-mode ablation as the theory-specific positive control.

Betti-Number Distribution and Analysis Design

The real-data row uses the manuscript values reported for monkey V1 windows: 46% full T^2 , 40% adjacent/partial signature, and 14% pooled as other. The remaining cells are an analysis template to populate from matched surrogate ensembles.

Table 2. Compact reporting table aligned with the current manuscript categories

Condition	Full T^2 (1, 2, 1)	Partial T^2 (1, 1, 1)	Other Betti configurations	Monte Carlo p vs real	Primary interpretation
Real monkey V1 LFP	46%	40%	14%	—	Observed population distribution.
Fourier phase shuffle	—	—	—	—	PSD-only null; amplitude distribution not preserved.
IAAFT	—	—	—	—	PSD + amplitude-distribution matched null.
Aperiodic-only 1/f	—	—	—	—	Tests whether broadband autocorrelation alone creates loops.
Envelope-preserving phase reset	—	—	—	—	Tests sustained phase continuity while preserving bursts.
One-mode ablation	—	—	—	—	Expected loss of one circular factor.

Important: The manuscript currently pools all non-(1,2,1) and non-(1,1,1) cases into “other.” Do not retroactively subdivide the 14% without re-running the classification. For the cleanest main-text comparison, retain these three pre-existing categories.

Table 3. Expanded Betti taxonomy for a new or supplementary analysis

Condition	T^2 (1,2,1)	Partial T^2 (1,1,1)	T^1 / circle (1,1,0)	Connected, no persistent loop (1,0,0)	Disconnected / noisy	Other
Real LFP	—	—	—	—	—	—
Phase shuffle	—	—	—	—	—	—
IAAFT	—	—	—	—	—	—
Aperiodic-only 1/f	—	—	—	—	—	—
Envelope-preserving phase reset	—	—	—	—	—	—
One-mode ablation	—	—	—	—	—	—

Use this expanded taxonomy only if the Betti-classification rule, persistence threshold, and handling of disconnected components are fixed before comparing conditions.

Statistical and Reconstruction Controls

These tables define how to compare the full Betti distribution and how to test whether DeepLagField recovers existing topology rather than manufacturing it.

Table 4. Matched surrogate-ensemble design

Step	Recommended implementation	Reason
1. Match windows	For every real recording window, generate one surrogate with the same length, sampling rate, and preprocessing history.	Preserves the hierarchical pairing between each real window and its null counterpart.
2. Build one complete null dataset	Repeat across all windows, then compute the aggregate Betti-category distribution.	Avoids treating many surrogates from one window as independent biological observations.
3. Repeat the ensemble	Repeat the full matched construction 200-500 times per surrogate class.	Produces a null distribution for the fraction of windows assigned to each Betti category.
4. Keep analysis fixed	Use identical delay τ , embedding dimension, point-cloud sampling, normalization, persistence threshold, and classification rules for real and surrogate signals.	Prevents the null result from depending on condition-specific tuning.
5. Compute Monte Carlo p	For full T^2 enrichment, use $p = [1 + \#\{q_s \geq q_{\text{real}}\}] / (S + 1)$, where q is the fraction classified as (1,2,1).	Provides an exact finite-ensemble significance estimate with the standard +1 correction.
6. Compare the full distribution	In addition to the T^2 fraction, report a paired/multinomial or permutation-based comparison of the complete category vector.	Shows whether the surrogate shifts topology specifically toward partial, one-loop, or non-loop configurations.

Table 5. Direct embedding versus model-reconstructed embedding

Analysis path	What it tests	Strongest expected result
Direct lag embedding of raw real and surrogate signals	Whether the topological signature already exists in the observed time series before DeepLagField reconstruction.	Real LFP shows greater T^2 enrichment than all matched surrogate classes.
DeepLagField reconstruction followed by lag embedding	Whether reconstruction improves detectability or instead induces toroidal topology.	DeepLagField increases detectability in real data but does not convert aperiodic-only or phase-reset surrogates into T^2 .
Difference-in-differences	Compare the reconstruction-induced change in T^2 fraction for real data versus each surrogate class.	A larger gain for real signals supports denoising/recovery of latent topology rather than generic model-induced torus formation.

Core causal narrative: aperiodic autocorrelation should not be sufficient for T^2 ; oscillatory bursts without sustained phase continuity should reduce complete T^2 detection; removing one mode should remove one circular factor; two sustained independent oscillatory phases should enrich (1,2,1).

Source note: Based on Figure 4 and Sections 2.2, 3.5, and Appendix B/J of the supplied NeuralFieldManifold manuscript. The manuscript reports monkey V1 Betti distributions of 46% (1,2,1), 40% (1,1,1), and 14% other.